Recap: Lecture 20

In Lecture 20, we

- Introduced BVPs (linear, 2nd-order)

$$f_2(x)y''(x) + f_1(x)y'(x) + f_0(x)y(x) = g(x), \qquad \begin{cases} a_1 y'(x_1) + b_1 y(x_1) & = c_1 \\ a_2 y'(x_2) + b_2\, y(x_2) & = c_2 \end{cases}$$

- A Dirichlet BC fixes the value of y at a boundary point and a Neumann BC fixes y'
- BVP solutions are solutions to homogeneous BVP ($g = c_1 = c_2 = 0$; form a solution space) + particular solution

Lecture 21

In Lecture 21, we will:

1. Introduce and classify PDEs
2. Introduce initial–boundary value problems
3. Introduce the 1D heat equation

The Laplacian (Vector Calculus revision)

The Laplacian is a linear transformation that appears in many common PDEs. It is denoted by ∇^2 and its representation changes depending on the coordinate system:

1. 2D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$$

2. 3D Cartesian

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

3. Cylindrical

$$\nabla^2 = \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \theta^2} + \frac{\partial^2}{\partial z^2}$$

PDEs and Classification

If there are multiple independent variables then the differential operators involve partial derivatives.

1. The *order* of a PDE is the order of the highest derivative.

2. A PDE is *linear* if it is a linear combination of partial derivatives of the unknown function, whose coefficients may only be functions of the independent variables, which equals a function of the independent variables. If it does not meet this criteria it is *non-linear*.

3. A PDE is *homogeneous* if each term contains the unknown function. It is *inhomogeneous* if there is a term which contains only the independent variables.

Partial differential equations (PDEs)

Examples of some PDEs include:

- Laplace's equation

$$\nabla^2 u = 0$$

- Heat/diffusion equation

$$\frac{\partial u}{\partial t} = \alpha \nabla^2 u$$

- Wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \nabla^2 u$$

- Schrödinger's equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \psi + V\psi$$

- Liouville equation

$$\nabla^2 u + \mathrm{e}^{\lambda u} = 0$$

- Burgers equation

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = \nu \frac{\partial^2 u}{\partial x^2}$$

Exercise 5.2: Classify the following PDEs

(a) $\nabla^2 f = x^2 y$, (b) $\dfrac{\partial^2 u}{\partial x \partial y} + \sin x \dfrac{\partial u}{\partial x} = 0$, (c) $u \dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} = \mathrm{e}^x$, (d) $\nabla^2 u = -\mathrm{e}^{\lambda u}$.

Initial conditions (ICs)

- General solutions of PDEs can contain *unknown functions*
  - General solutions of ODEs have unknown constants, which we try to specify with initial or boundary conditions.
- If the PDE is first order in time, we usually constrain the unknown function to be some specified function, $u(\boldsymbol{x}, t)$ at the initial time, i.e., $u(\boldsymbol{x}, 0) = f(\boldsymbol{x})$.
- If it is second order in time, we impose an additional constraint on the *time* derivative of the unknown function, i.e., $\partial_t u(\boldsymbol{x}, 0) = g(x)$.

Boundary conditions (BCs)

- Boundary conditions are constraints imposed at specified locations in space for all times after the initial condition.
- Let D be a bounded spatial domain, then boundary conditions are imposed on the domain's boundary, ∂D.
- Two most common boundary conditions are:
  1. *Dirichlet BCs* fix the value of the unknown function on the boundary, i.e., $u(\boldsymbol{x}, t) = f(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$.
  2. *Neumann BCs* fix the value of the normal derivative at the boundary, i.e., $\partial_{\boldsymbol{n}} u(\boldsymbol{x}, t) = g(\boldsymbol{x}, t)$ where $\boldsymbol{x} \in \partial D$ and $\boldsymbol{n}$ is the boundary's outwards unit normal.

A PDE with ICs and BCs is called an *initial-boundary value problem* (IBVP)

What will we solve?

1. The heat/diffusion equation in one and two-dimensions, in cartesian and axisymmetric polar coordinates (α is called the thermal diffusivity)

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}, \qquad \frac{\partial u}{\partial t} = \alpha \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \qquad \text{and} \qquad \frac{\partial u}{\partial t} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right).$$

2. The wave equation in one and two-dimensions, in cartesian and axisymmetric polar coordinates (c is called the wave speed)

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}, \qquad \frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \qquad \text{and} \qquad \frac{\partial^2 u}{\partial t^2} = \frac{c^2}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right)$$

3. The two-dimensional Laplace equation in cartesian and polar coordinates

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \qquad \text{and} \qquad \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$$

*We will also solve some inhomogeneous examples

Classification of second-order PDEs

Suppose the PDE is of the form

$$A\,u_{xx} + B\,u_{xy} + C\,u_{yy} + \text{lower-order terms} = 0$$

The PDE is:

- *Elliptic* if $B^2 - 4\,A\,C < 0$
- *Parabolic* if $B^2 - 4\,A\,C = 0$
- *hyperbolic* if $B^2 - 4\,A\,C > 0$

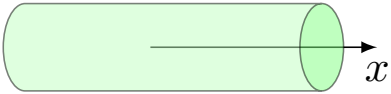
Case study 7: Heat conduction in a quartz rod

- **Quartz** is unique due to its special mechanical, electrical, thermal, chemical and optical properties.
- It has an exceptional thermal stability. For example, a piece of quartz will not break if heated to a temperature of 1100°C, then suddenly cooled down to 20°C.
- **WHY CARE? Well this is super important if you have ambitions of becoming 007 since your job will require you to run through burning buildings and cars – all in a day's work!**

So .. this is why Bond only wears Quartz watches.

The 1D heat equation

- Consider a straight rod of uniform cross section and homogeneous matrial (quartz)



- Let the temperature along the rod be $u(x,t)$, where x is axial location and t is time
- Fourier's law: Heat flow $= -\kappa \dfrac{\partial u}{\partial x}$ where $\kappa =$ thermal conductivity
- Rate of change in heat $= c\,\rho \dfrac{\partial u}{\partial t}$ where $c =$ specific heat capacity, $\rho =$ density
- Conservation law: $c\,\rho \dfrac{\partial u}{\partial t} = -\dfrac{\partial}{\partial x}\left(-\kappa \dfrac{\partial u}{\partial x}\right) \quad \Rightarrow \quad \dfrac{\partial u}{\partial t} = \alpha \dfrac{\partial^2 u}{\partial x^2}$

  where $\alpha = \kappa / c\,\rho =$ thermal diffusivity

Thermal diffusivity

Material	α [$\mathrm{cm^2\,s^{-1}}$]
Gold	1.27
Copper	1.11
Silicon	8.8×10^{-1}
Aluminium	8.4×10^{-1}
Iron	2.3×10^{-1}
Air	1.9×10^{-1}
Steel	1.2×10^{-1}
Quartz	1.4×10^{-2}
Brick	3.8×10^{-3}
Water	1.4×10^{-3}
Wood	8.2×10^{-4}